

```

create database retail_sales_analysis;
use retail_sales_analysis;

create table customers(
customer_id int primary key auto_increment,
customer_name varchar(225),
city varchar(225),
date date);

create table products(
product_id int primary key auto_increment,
product_name varchar(225),
category varchar(225),
price decimal(9,2));

create table orders(
order_id int primary key auto_increment,
customer_id int,
product_id int,
quantity int not null,
oreder_date timestamp
);

CREATE TABLE PRODUCTS_CUSTOMERS(
    PRODUCT_ID INT,
    CUSTOMER_ID INT,
    QUANTITY INT,

    CONSTRAINT fk_pc_product
        FOREIGN KEY (PRODUCT_ID)
        REFERENCES PRODUCTS(PRODUCT_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    CONSTRAINT fk_pc_customer
        FOREIGN KEY (CUSTOMER_ID)
        REFERENCES CUSTOMERS(CUSTOMER_ID)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    PRIMARY KEY (PRODUCT_ID, CUSTOMER_ID)
);

SELECT * FROM PRODUCTS_CUSTOMERS;

INSERT INTO CUSTOMERS VALUES
(1, 'rahul sharma','MUMBAI', '2020-02-03'),
(2, 'Priya Patel', 'PUNE', '2021-03-02'),
(3, 'Amit Verma', 'Delhi', '2020-04-05'),
(4, 'Sneha Gupta', 'Pune', '2023-08-23'),
(5, 'Rohan Mehta', 'Bangalore', '2022-06-20'),
(6, 'Anjali jadhav', 'Lucknow', '2021-04-23'),
(7, 'Vikram jadhav', 'Jaipur', '2022-02-19'),
(8, 'Neha mulani', 'Nagpur', '2023-02-02'),
(9, 'Karan Malhotra', 'Chandigarh', '2024-04-24'),
(10, 'Pooja more', 'Surat', '2025-12-12'),
(11, 'ankita gaikwad', 'panchwad', '2026-02-02'),
(12, 'siddhi mohite', 'karad', '2022-03-03'),
(13, 'srushti banker', 'wai', '2023-04-04'),
(14, 'triveni nalawade', 'satara', '2022-04-06'),
(15, 'shivali dhmal', 'kokan', '2021-09-09'),

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(16, 'arpita lohar', 'patan', '2020-09-09'),
(17, 'sanjivani kare', 'wadhe', '2019-09-29'),
(18, 'sanajan jedhe', 'wai', '2018-08-08'),
(19, 'arpita chavan', 'wai', '2025-09-08'),
(20, 'anjali more', 'satara', '2026-08-09');
```

insert into products values

```
(1, 'Laptop', 'Electronics', 55000),
(2, 'Smartphone', 'Electronics', 30000),
(3, 'Headphones', 'Electronics', 2500),
(4, 'Refrigerator', 'Home Appliances', 42000),
(5, 'Washing Machine', 'Home Appliances', 38000),
(6, 'Microwave Oven', 'Home Appliances', 12000),
(7, 'Office Chair', 'Furniture', 7000),
(8, 'Study Table', 'Furniture', 9500),
(9, 'LED TV', 'Electronics', 45000),
(10, 'Bluetooth Speaker', 'Electronics', 3500),
(11, 'Air Conditioner', 'Home Appliances', 52000),
(12, 'Ceiling Fan', 'Home Appliances', 2800),
(13, 'Gaming Mouse', 'Electronics', 1500),
(14, 'Keyboard', 'Electronics', 1200),
(15, 'Water Purifier', 'Home Appliances', 18000),
(16, 'Dining Table', 'Furniture', 25000),
(17, 'Bed Mattress', 'Furniture', 14000),
(18, 'Smart Watch', 'Electronics', 8000),
(19, 'Tablet', 'Electronics', 27000),
(20, 'Power Bank', 'Electronics', 2000);
```

TRUNCATE TABLE orders;

INSERT INTO orders VALUES

```
(1,1,1,1,'2023-01-05'),
(2,2,2,2,'2023-01-06'),
(3,3,3,3,'2023-01-07'),
(4,4,4,4,'2023-01-08'),
(5,5,5,5,'2023-01-09'),
(6,6,6,6,'2023-01-10'),
(7,7,7,7,'2023-01-11'),
(8,8,8,8,'2023-01-12'),
(9,9,9,9,'2023-01-13'),
(10,10,10,10,'2023-01-14'),
(11,11,11,11,'2023-01-15'),
(12,12,12,12,'2023-01-16'),
(13,13,13,13,'2023-01-17'),
(14,14,14,14,'2023-01-18'),
(15,15,15,15,'2023-01-19'),
(16,16,16,16,'2023-01-20'),
(17,17,17,17,'2023-01-21'),
(18,18,18,18,'2023-01-22'),
(19,19,19,19,'2023-01-23'),
(20,20,20,20,'2023-01-24');
```

-- which products generate the most revenue...?

```
SELECT p.product_name,
SUM(o.quantity * p.price) AS revenue
FROM orders o
JOIN products p
ON o.product_id = p.product_id
GROUP BY p.product_name
ORDER BY revenue DESC;
```

-- who are the top customers..?

```
SELECT c.customer_name,  
SUM(o.quantity * p.price) AS total_spent  
FROM orders o  
JOIN customers c  
ON o.customer_id = c.customer_id  
JOIN products p  
ON o.product_id = p.product_id  
GROUP BY c.customer_name  
ORDER BY total_spent DESC;
```

-- what are the monthly sales trends...?

```
select month(oreder_date) as month,  
sum(quantity) as monthly_sales  
from orders  
group by month(oreder_date);
```

```
select * from orders;
```

-- which category performs best...?*

```
select p.category,  
sum(o.quantity * p.price) as best_perform  
from orders o  
join products p  
on p.product_id = o.product_id  
group by p.category  
order by best_perform desc;
```